

INTERNSHIP REPORT PRESENTATION

Ethical Hacking

Cybersecurity Internship

Authorized security testing • Vulnerability assessment • Defensive security

Anshu Raj

Roll No. 2410030252 | B.Tech CSE

June – August 2026

Candidate Profile



**Anshu
Raj**

B.Tech – Computer Science & Engineering



Roll Number

2410030252



Institute

IILM University, Greater Noida, U.P.



Programme

B.Tech CSE, Batch 2024–2028



Internship Domain

Ethical Hacking / Cybersecurity



Duration

8 Weeks | June – August 2026

ABOUT THE PROGRAM

Ethical Hacking & Cybersecurity

The internship focuses on practical cybersecurity concepts: understanding attack surfaces, identifying vulnerabilities, performing controlled security testing, and translating findings into defensive improvements.

AUTHORIZED TESTING

Security testing is performed only with explicit permission and within a defined scope.

VULNERABILITY ASSESSMENT

Identify weaknesses, understand their impact, and prioritize remediation.

DEFENSIVE SECURITY

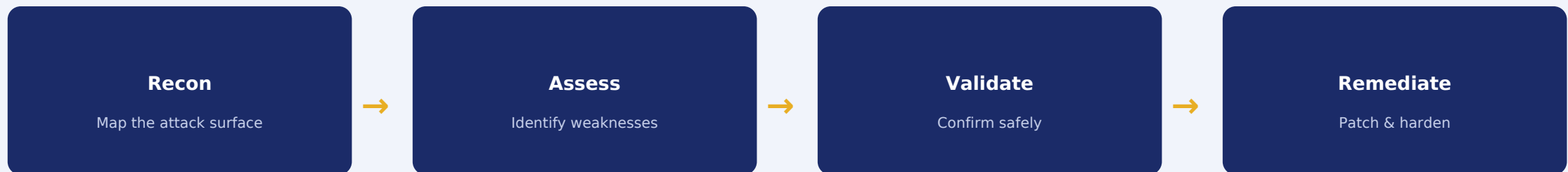
Use findings to improve configurations, access controls, patching, monitoring, and awareness.

PROBLEM STATEMENT

Bridging Theory and Applied Cybersecurity

Many students understand networking, operating systems, and security concepts theoretically, but have limited structured practice applying them to realistic security scenarios. The internship addresses this gap through a controlled ethical-hacking workflow.

This internship addressed that gap — practically learning the full security workflow:



What the Internship Set Out to Achieve

Reconnaissance

Understand authorized information gathering and attack-surface mapping.

Network Security

Learn ports, services, protocols, firewalls, and secure configurations.

Web Security

Recognize common application weaknesses and safe validation methods.

Vulnerability Analysis

Assess findings, understand risk, and prioritize remediation.

Security Tools

Use ethical-hacking tools responsibly in controlled lab environments.

Reporting

Document evidence, impact, and recommended fixes clearly.

The Toolkit Behind the Internship

Kali Linux

Security-focused operating system

Nmap

Network discovery & service enumeration

Wireshark

Network traffic analysis

Burp Suite

Web application security testing

Metasploit

Controlled vulnerability validation

OWASP ZAP

Web security testing

Python

Automation & security scripting

Linux CLI

System administration & investigation

From Target Asset to Verified Remediation



The 8-Week Learning Journey

1 WEEK

Security Fundamentals

Networking, Linux, CIA triad & authentication

2 WEEK

Reconnaissance

Asset discovery and service identification

3 WEEK

Vulnerability Assessment

Identify, classify and prioritize findings

4 WEEK

Web Security

Requests, sessions, input validation & OWASP concepts

5 WEEK

Controlled Testing

Validate findings within authorized scope

6 WEEK

Defensive Hardening

Patching, configuration and access controls

7 WEEK

Incident Awareness

Logs, indicators and response basics

8 WEEK

Reporting

Evidence, risk, remediation and retesting

EXPECTED OUTCOMES

Skills & Capabilities Gained

- ✓ Understand the ethical-hacking lifecycle and security testing mindset.
- ✓ Perform basic reconnaissance in authorized environments.
- ✓ Analyze network traffic and protocol-level risks.
- ✓ Assess web applications using safe, controlled techniques.
- ✓ Interpret vulnerability findings and prioritize risk.
- ✓ Use security tools responsibly in labs and authorized scopes.
- ✓ Produce professional security reports with evidence and remediation.
- ✓ Develop a defensive mindset focused on prevention and verification.

Successfully Completed

Student

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Ethical Hacking

Internship focused on ethical hacking, vulnerability assessment, controlled security testing, and defensive security practices.

Responsible Security • Practical Learning

THANK YOU

Ethical Hacking

Cybersecurity • Defensive Thinking • Responsible Testing

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